25901130 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBOT200 - BIOFERTILIZERS AND BIOCONTROL AGENTS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any 10 questions

Each question carries 1 marks

1. List any two objectives of minimal cultivation. [K] / [CO1]
   Saves time, labor, fuel, and energy
   Prevents soil erosion and loss of soil fertility
   Keeps the soil structure strong for better plant growth
2. What are bio-fertilizers? [K] / [CO2]
   Natural fertilizers made from living microorganisms that help plants grow by increasing nutrients in the soil.
3. Why is Rhizobium inoculation ineffective for cereal crops like wheat? [K] / [CO2]
   Rhizobium bacteria form nodules only in the roots of leguminous plants, not in cereals.
4. Which specific gene in Bt is responsible for producing insecticidal protein crystals? [K] / [CO3]
   Cry gene
5. How biopesticides protect plants ? [U] / [CO3]
   Biopesticides protect plants by controlling harmful insects and plant diseases. They are made from natural organisms or substances and are safe for the environment.
6. What is microbial inoculation in composting? [K] / [CO5]
   Microbial inoculation in composting is the process of adding useful microorganisms to compost. These microbes help break down organic waste faster and improve the quality of compost
7. How active ingredients in bioformulations influence their activity? [K] / [CO4]
   Active ingredients in bioformulations are the microbial organisms which possess the real insecticidal activity. They determine how effectively the biopesticide can control pests or diseases. They work by attacking or inhibiting the harmful organisms.

8. Name one benefit of using biofertilizers. [K] / [CO6
   They reduce the need for chemical fertilizers.
   They are safe for the environment
   They improve soil fertility
9. Name a fungi that is used as a biocontrol agent. [K] / [CO4]
   Trichoderma
10. Name one enzyme involved in composting. [K] / [CO5]
    Cellulase or pectinase
11. Name any two nitrogen-fixing microorganisms used as bio-fertilizers. [K] / [CO2]
    Rhizobium/anabena/nostoc/azetobacter etc
12. What is the function of fungi in compost? [K] / [CO5]
    Fungi break down complex materials like cellulose and lignin in organic waste. This helps convert plant residues into simple nutrients in compost

Part B

Short Essay

Answer any 6 questions

Each question carries 5 marks

13. Give a brief account on sustainable agriculture. [U] / [CO1]
    It is a method of farming that aims to produce food while protecting the environment and conserving natural resources. It helps maintain soil fertility and prevents soil erosion. Sustainable agriculture reduces the use of chemical fertilizers and pesticides by using natural alternatives. It promotes the use of biofertilizers and biopesticides. This method also conserves water and improves soil health. It helps protect biodiversity and maintains ecological balance. Sustainable agriculture supports long-term food production for future generations. It also helps farmers reduce production costs and maintain healthy crops.
14. Do we need to promote the production and use of biopesticides? Justify your answer. [An] / [CO3]
    Yes, we need to promote the production and use of biopesticides. Biopesticides are safe for the environment and cause less pollution. They are not harmful to humans, animals, and beneficial insects. They help control pests naturally. They reduce the use of harmful chemical pesticides. They help maintain ecological balance. They improve soil health. They also support sustainable agriculture.
15. How does Trichoderma contribute to soil health and the decomposition of organic matter? [U] / [CO3]
    Trichoderma acts as a biocontrol agent (controls harmful pathogens) in the following ways:
    - Competition: Competes with harmful fungi for space and nutrients, preventing their growth
    - Mycoparasitism: Directly attacks and feeds on pathogenic fungi by coiling around and breaking their cell walls
    - Antibiosis: Produces antibiotics and toxic substances that inhibit or kill plant pathogens
    - Enzyme production: Secretes enzymes like chitinase and glucanase that destroy the cell walls of harmful fungi
    - Induced resistance: Stimulates the plant's natural defense system, making it more resistant to diseases
    - Rapid growth: Grows quickly and colonizes the root zone, leaving little chance for pathogens to establish
16. Write an account on bioformulations used for the uptake of Phosphorus[K] / [CO4]
    Bioformulations used for the uptake of phosphorus are biological products that contain beneficial microorganisms capable of increasing the availability of phosphorus in soil for plant absorption. Phosphorus is an essential nutrient required for root development, , but much of it in soil exists in insoluble forms that plants cannot directly use.
    Active component of these bioformulations include phosphate-solubilizing microorganisms (PSMs) such as species of *Bacillus*, *Pseudomonas*, and fungi like *Aspergillus* and *Penicillium*. These microbes secrete organic acids (such as citric, oxalic, and gluconic acids) that convert insoluble phosphates into soluble forms like orthophosphates, making them available to plants. Some microorganisms also produce enzymes like phosphatases that mineralize organic phosphorus

compounds.
Mycorrhizal fungi(VAM fungi), also form symbiotic associations with plant roots. Their hyphal network extends into the soil, increasing the surface area for absorption and enhancing phosphorus uptake from distant soil zones.
Bioformulations are prepared by mixing these beneficial microbes with carriers such as peat, lignite, or compost, and are applied as seed treatment, soil application, or root dipping. Their use improves phosphorus use efficiency, reduces dependence on chemical fertilizers, enhances soil fertility, and supports sustainable agriculture.

17. Write a comparative account on wettable powder and dust type of bioformulations[U] / [CO4]
Wettable powder (WP) bioformulations are dry, fine powders containing beneficial microorganisms along with carriers and wetting agents. They are mixed with water before application and applied as sprays, seed treatments, or soil drenches. Dust type bioformulations, on the other hand, are dry formulations where microbial cultures are mixed with inert carriers like talc or ash and applied directly in powder form. They are mainly used for seed dressing or soil application without mixing in water.
- Wettable powders require mixing with water; dust formulations are applied directly.
- Wettable powders provide uniform application; dust types may be uneven.
- Wettable powders adhere better to plant surfaces; dust types have lower adhesion.
- Dust formulations are easier and quicker to apply; wettable powders need preparation.
- Wettable powders are generally more efficient; dust types are less precise in application.

18. Write briefly about the advantages of using Trichoderma instead of chemical pesticides. [U] / [CO6]
1. It is eco-friendly and does not pollute soil or water.
2. It is safe for humans, animals, and beneficial organisms.
3. It acts as a natural biocontrol agent against many plant pathogens.
4. It improves soil health by enhancing microbial activity.
5. It promotes plant growth and root development.
6. It reduces dependence on harmful chemical pesticides.
7. It helps in long-term disease suppression in soil.
8. It is biodegradable and leaves no toxic residues on crops.

19. Give a brief account on potash-solubilising bio-fertilizers. [U] / [CO2]
Potash-solubilising biofertilizers (KSB) are microbial formulations that help in making potassium (potash) available to plants from insoluble sources in the soil. Potassium is an essential nutrient required for overall plant growth, but a large portion of soil potassium remains in unavailable mineral forms.
These biofertilizers contain beneficial microorganisms such as *Bacillus sp*, and certain species of *Pseudomonas*. They solubilize potassium-bearing minerals by producing organic acids and other compounds that release potassium into a soluble form that plants can absorb.
Potash-solubilising biofertilizers are usually applied as seed treatment, soil application, or mixed with compost. Their use enhances potassium uptake, improves crop yield and quality, and increases resistance to stress conditions such as drought and disease.

20. Describe biodegradable waste and its importance in compost production. [U] / [CO5]
Biodegradable waste refers to organic materials that can be broken down naturally by microorganisms like bacteria and fungi into simpler substances. Examples include kitchen waste (vegetable peels, fruit scraps), garden waste (leaves, grass), and animal waste.
Importance in compost production:
- It serves as the main raw material for making compost.
- Microorganisms decompose biodegradable waste into nutrient-rich humus.
- It improves soil fertility by adding essential nutrients.
- It enhances soil structure and water-holding capacity.
- It promotes the growth of beneficial soil microbes.
- It reduces the amount of waste sent to landfills.
- It is an eco-friendly way of waste management.
- It supports sustainable agriculture by reducing the need for chemical fertilizers.

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. Discuss ectomycorrhiza as a biofertilizer. [U] / [CO2]

Ectomycorrhiza is a beneficial relationship between certain fungi and the roots of plants, especially trees like pine, oak, and eucalyptus. In this association, the fungus forms a sheath (covering) around the roots and helps the plant absorb nutrients from the soil, while the plant provides food (sugars) to the fungus.

Ectomycorrhiza plays an important role in improving plant growth naturally without using chemicals. It plays a key role in sustainable agriculture and forest management.

Functions and role:

- The fungal threads spread into the soil and increase the root's ability to absorb nutrients.
- It helps in better uptake of important nutrients like phosphorus and nitrogen.
- It also improves water absorption, especially in dry conditions.
- The fungus protects plant roots from harmful pathogens (disease-causing organisms).

Advantages:

- Improves nutrient uptake, especially phosphorus.
- Enhances plant growth and survival.
- Increases resistance to diseases and environmental stress.
- Improves soil structure and fertility.
- Reduces the need for chemical fertilizers.
- Helps plants grow well even in poor soils.
- Eco-friendly and safe for the environment.

22. Explain the isolation, mass multiplication, and formulation of Rhizobium as a biocontrol agent. [U] / [CO6]

Rhizobium is a beneficial soil bacterium found in the root nodules of legume plants like peas, beans, and groundnut. It converts atmospheric nitrogen into a usable form for plants. It also acts as a biocontrol agent by improving plant health and suppressing harmful microbes.

Isolation of Rhizobium:

Isolation is the process of obtaining a pure strain. Healthy pink root nodules are collected from legume plants. The nodules are washed and surface sterilized to remove unwanted microorganisms. They are then crushed and spread on a selective medium such as Yeast Extract Mannitol Agar (YEMA). After incubation, bacterial colonies appear and are purified. The best strain is selected based on its ability to form nodules and fix nitrogen.

Mass Multiplication:

Mass multiplication is the large-scale production of *Rhizobium*. A nutrient medium like Yeast Extract Mannitol (YEM) broth is prepared and sterilized. The selected culture is inoculated into the broth and incubated at 28–30°C. The bacteria multiply rapidly and reach a high population (10 –10 cells/ml). The culture is then ready for further processing.

Formulation of Rhizobium Biofertilizer:

In formulation, the bacterial culture is mixed with a carrier material such as peat or lignite. The carrier is first powdered, sterilized, and moisture is adjusted to about 40–50%. The broth culture is mixed uniformly with the carrier and kept for 2–3 days (curing). The final product is packed in polythene bags and stored at 4–15°C. The shelf life is about 6 months.

Role as a Biocontrol Agent:

Rhizobium competes with harmful microbes in the soil, improves plant growth, enhances soil fertility, and increases plant resistance to diseases. It reduces the use of chemical fertilizers and pesticides.